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SMART GARBAGE MANAGEMENT SYSTEM

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Abstract:

Every single day, garbage is always produced and sometimes, due to the imbalance between the high volume produced and the garbage volume transported to the landfill; it then leads to the buildup. The current garbage collection system includes routine garbage trucks doing rounds daily or weekly, which not only doesn't cover every zone of the city but it's a completely inefficient use of government resources. To prevent any negative impact on the environment, a system is needed to support the waste management process. This project consists of two parts: a portable application using an android smartphone and a monitoring application. The user can upload garbage images, and the location of a particular area and share the comment by using the Android mobile application. By using a monitoring application, the admin collects all the information of the user's details from the android mobile app. If the details are considered valid, they will give rewards to users and take necessary actions. This project provides a cost-effective mobile system for the government to utilize available resources to efficiently manage the overwhelming amounts of garbage collected each day. The server will collect the data and store the information in a database. If the information is considered valid, then they will take action by the admin. The admin rewards the person who is sending the valid information.

Keywords: Admin, Android app, Cost-effective, Garbage Images, Monitoring Application, User

I.Introduction:

This design will reduce the threat of hygienic conditions of the smart megacity. With adding population in our country, urbanization has increased which has led to the generation of waste. Tons of trash are generated every time. Ten million tons of scrap is generated in the metropolises. This leads to deterioration of public health, environmental pollution, and impact on quality of life of the citizens. therefore, waste operation has been a pivotal issue. former exploration in the area of technology related to Smart Garbage Management involves the use of Zigbee and GSM technology for connectivity. before technology includes RFID technology to identify markers attached to objects. A combination of the below technologies has been used for connectivity purposes. There's a space in exploration to efficiently warn the authorities in an optimized way to maintain clean terrain. The system introduced uses more advanced wireless technology and handy operations for farther advancements. The scrap full condition is automatically detected with the help of threshold value set and the information is transmitted to the concerned authorities to initiate the cleaning operations. The medium used to detect the caddy helps in minimizing the time needed for performing the cleaning process.

II.Problem Statement:

A rise in population and economy, which results in increase in amount of wastes generated. This puts pressure on garbage management facilities, which are already in short supply. Increased complexity of the waste stream because of urbanization and industrialization. The complexity of the waste stream directly affects the complexity of its management, which is compounded when hazardous waste mixes the general waste. Limited understanding of the main waste flows and national waste balance because the submission of waste data is not obligatory, and where data is availability, it is often unreliable and contradictory. Growing pressure on outdated garbage management infrastructure, with declining levels of capital investment and maintenance. Garbage management suffers from pervasive under – pricing, which means that the costs of garbage management are not fully appreciated by consumers and industry, and waste disposal is preferred over other options. Few waste treatment options are available to manage waste and so they are more expensive than landfill costs.

III.Literature Survey:

[1]Bhanu G. Priya stated that most serious environmental concerns at present have been solid waste management which contributes to disturbing the balance of the environment and that has adverse effects on the health of society. Detection, monitoring and management of urban wastes are one of the primary problems of the present era. [2] K. Aditya Rao, Dr. Ganesh D, put forth the statement that smart cities integrate multiple mobile or web solutions to build a comfortable human habitation. One of these solutions is to provide an environmentally friendly, efficient and effective garbage management system. The current garbage collection system includes routine garbage trucks doing rounds daily or weekly, which not only doesn't cover every zone of the city but it's completely inefficient use of government resources. [3]Dr. Raveesh Agarwal, Mona Chaudhary, Jayveer Singh proposed that the objective of writing this paper is to study the current practices related to the various waste management initiatives taken in India for human wellbeing. The other purpose is to provide some suggestions and recommendations to improve the waste management practices in Indian towns. This paper is based on secondary research. [4]Yunan He, Chen Yang, Xiao-Feng Li proposed that Google Android is a popular software stack for smart phones, where the user experience is critical to its success. The pause time of its garbage collection in Dalvik VM should not be too long to stutter the game animation or webpage scrolling. Generational collection or concurrent collection can be the effective approaches to reducing GC pause time. As of version 2.2, Android implements a non-generational stop the-world (STW) mark-sweep algorithm.

IV.Proposed System:

From mobile application, the public directly able to share information about the garbage overflow to the admin. The server will collect the data and store the information in a database. If the information is considered as valid, then the action will be taken by the admin. The admin rewards the person who is sending the valid information. The location of the garbage is tracked with the help of the GPS from the location where the user uploads the picture. Then it will enable to reduce the overall time in tracking and disposing the dustbin on time. This information essentially helps the respective department to take care of garbage present in the society. Based on the data collected, garbage trucks can be given routes generated through GPS system to efficiently route through the particular garbage bin. If they are not taking proper action, then the public will send the information through higher officers. On following this method, the garbage could be cleaned in an efficient way.

V.Architecture:

Two modules will be present namely User, Admin If you are a user, click on the User Module. Login page will be displayed on the screen. Enter your login credentials if you are an existing user. If you are a new user, then click on create an account option. Create a new account using your information and then login to your account. In the Home screen three modules will be present namely Garbage System, Profile and Send Complaints. To post a complaint select the send Complaints module and add the description about your complaint and add the related photos and upload it. After successfully uploading your complaint. The history of your complaints can be viewed on the Garbage System Module. The Username, Mobile number, Email-id, City, and Rating history will be available in the profile module. The User will receive an SMS if his/her complaint is accepted by the Admin. If you are an Admin select the admin module. Login with your credentials in the Admin Login Page. It will direct you to the request page. The various requests received from the users will be displayed. If the Admin selects the desired request, it will display the Requested user details, their uploaded complaint picture, the GPS location of the posted complaint. The admin verifies the queries posted if he considers them valid one, he accepts them or otherwise he will cancel them. The status of their request if accepted a message will be sent to the desired requested user.

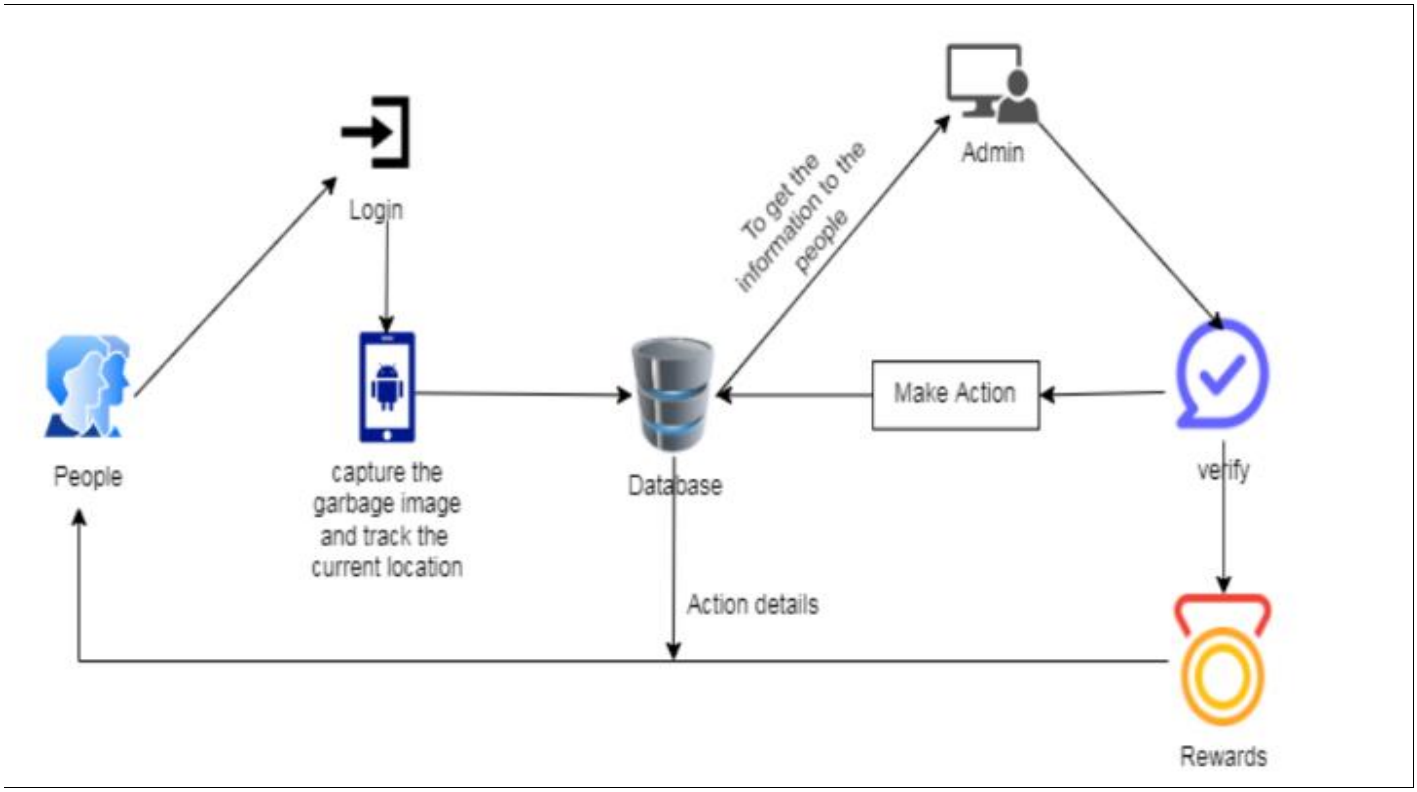


Fig: 5.1 - Architecture Diagram

VI.Implementation:

Our Smart Garbage Management app was designed and developed using XML and JAVA the standard language for Android development. The app was created using Android Studio. We divided the whole development process into modules and modules are further divided into submodules. XML is used to create the backgrounds and buttons. For the backend, Firebase has been used for authentication, storage and database. To test the working model for our application we firstly used Android Emulator then for Alpha Testing we activated 'Developer Options' in mobile phone and tested it.

VII.Results:

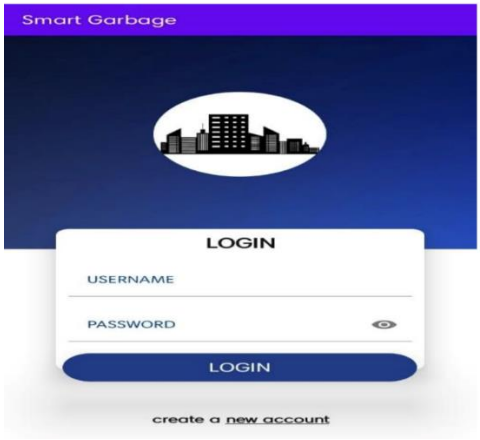


Fig : 7.1 - Login Page

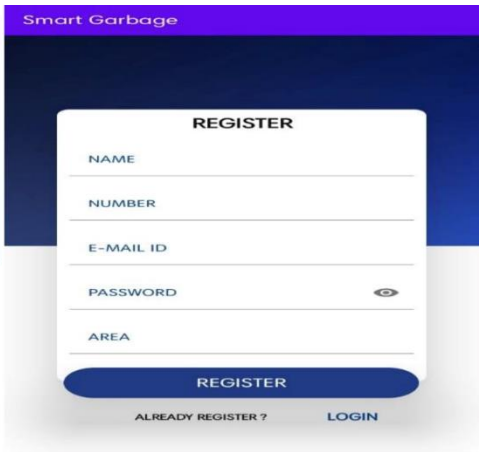


Fig : 7.2 - Register Page

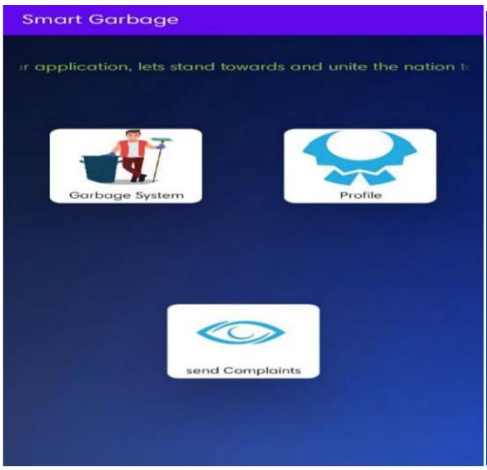


Fig : 7.3 - User Home Page



Fig : 7.4 - Admin Login Page



Fig : 7.5 - Profile page

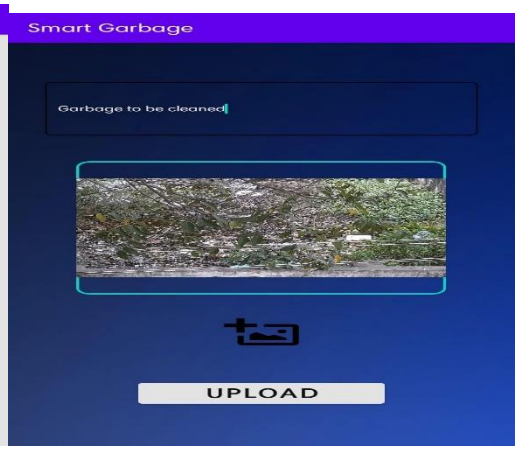


Fig : 7.6 - Complaints upload page

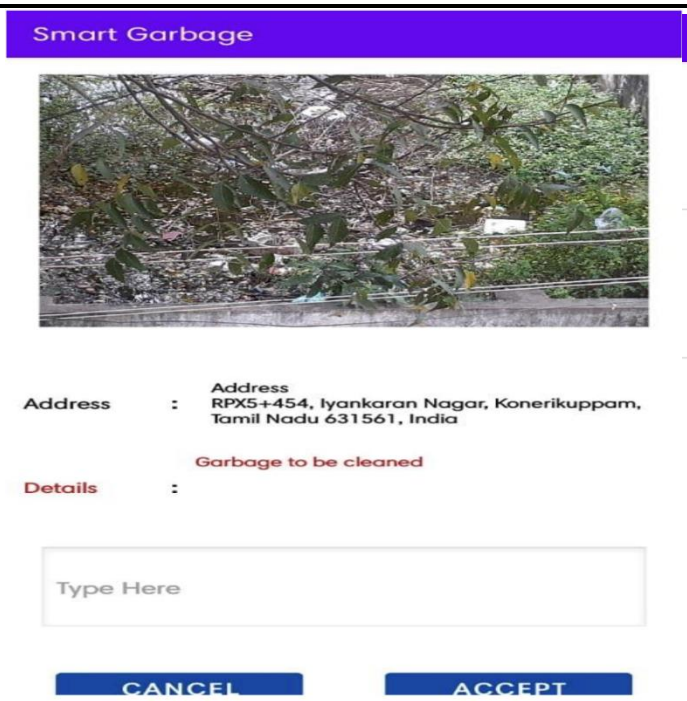


Fig : 7.7 - Complaint's Accept/Cancel Page

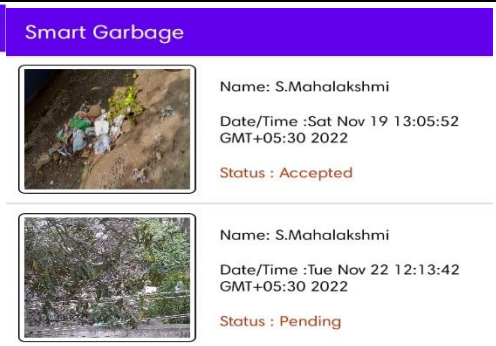


Fig : 7.8 - Complaint Status Page

VIII.Conclusion:

This paper proposes a system that makes sure that cleaning of garbage bins is done. The system uses the android application and Wi-Fi module. This further reduces operational time by alerting the concerned authorities via an Android application. This system also makes use of GPS technology in the Android application to alert the nearest employee and hence reducing the time for the dustbin cleaning process.

IX. Future Enhancement:

In the future, this project may be implemented with some more algorithms for checking the best accuracy score better than the current method. We will be testing the model with different types of datasets. To gain a better understanding, we will use the data visualization process to compare the various models.

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